

B.E./Insem/Oct-80

**B.E. (Computer Engineering) (Semester - I)
DESIGN & ANALYSIS OF ALGORITHMS
(2012 Pattern)**

Time :1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q3 or Q.4, Q.5 or Q.6.*
- 2) *Draw neat diagram whenever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) Solve the following recurrence relation using master theorem [6]

i) $T(n) = T\left(\frac{n}{2}\right) + \theta(1)$

ii) $T(n) = 4T(n/2) + n^2$

b) Explain accounting method of Amortized Analysis with Stack example[4]

OR

Q2) a) Write an algorithm for Recursive Binary Search. What is the Time Complexity for Successful Search and Unsuccessful Search? [6]

b) Define asymptotic notations. Explain their significance in analyzing algorithms. [4]

Q3) a) Write an algorithm to solve knapsack problem using greedy strategy. State its time complexity. [6]

b) Explain use of dynamic programming to compute a binomial coefficient. State its time complexity. [4]

OR

Q4) a) Using job scheduling deadlines find the optimal schedule for following jobs with $n = 6$. Profits:(P1,P2,P3,P4,P5,P6) = (20,15,10,7,5,3) and deadline: (d1,d2,d3,d4,d5,d6) = (3,1,1,3,1,3) [6]

b) Write control abstraction for greedy strategy. [4]

P.T.O.

- Q5)** a) Give the Definition of Implicit Constraint and Explicit Constraint. State and Explain both the constraint using 8 Queen Example [6]
b) State the control abstraction for FIFO based Branch and Bound method. [4]

OR

- Q6)** a) Write an algorithm to solve N-Queens problem. [8]
b) Define the following terms with respect to branch and bound strategy (any 2):
i) State space tree, ii) Live node, iii) E-node, iv) Answer node [2]

